

ActInf MorphStream 001.1 ~ David Kappel and Sarah Hamburg

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Hello everybody welcome it is September 26th 2023 we are here kicking off a new stream series at the Active inference Institute this is the morph stream 1.1 today we have David kappel and also this section and streams facilitated by Sarah Hamburg we're going to have a overview first presented by Sarah then David will share some work on neuromorphic Computing and then we'll have some time to discuss so thank you both for joining and Sarah to you for the first presentation and also to introduce yourself if you'd like yes that's a good idea thank you very much Daniel um so my name is Sarah um I'm a neuroscientist specializing intelligence currently working in the field of neuromorphic computing um at chef field Hal in the UK so I'm going to give you a high level overview of what neuromorphic Computing is and before we hear David's exciting talk in the first edition of this new series um just to let you know if you're watching on double time in the future I talk quite fast so you might not want to watch me on double time so this QR code um I put here will take you to a paper which I thought was a really nice introduction to the field um but neic Computing can be defined as Computing systems that are designed to mimic the structure and function of the nervous system so this doesn't have to be the human nervous system the field actually takes inspiration from all sorts of animals and insects um although the definitions online don't necessarily acknowledge that so some people are quite open with what constitutes neuromorphic while maybe others would prefer neuromorphic was reserved for Hardware instantiations of biological like neurons um which are sometimes referred to as non-run computers and I think that's what the paper that the QR code um will refers to it as in as their definition um so what I think is really interesting is a little bit of the context so like our current vuman computer architecture was also inspired by Neuroscience and particularly the Mull and pits 43 Neuron model um inspired Von y's first draft in 1945 so Neuroscience has a long history of um inspiring computer science and this also includes reinforcement learning which is based on theories about learning decision- making from behavioral psychology based on rewards and

punishments and also heian learning principles of cells that fire together wire together um from 49 became foundational for unsupervised learning so first of all um hang on one second so um in order to understand the why of neuromorphic computing I really wanted to explain what's so great about the brain so uh here's some inspiration for light bulbs so I'm going to ask you a question I just want you to think about it for a second in terms of light bulbs how much energy do you think the brain uses do you think it's more or less energy than the bulbs lighting the room that you're in if you're in the future by all means PA this pause this if if you want to do some in-depth calculations but I'm going to skip to the answer the answer is here in the pink circle so it's 20 watts so that's the equivalent of one modern day energy efficient light bulb so that's probably what's above me now basically in my room here um this QR code should take you to quite an interesting paper on power consumption in the brain if you're interested in that so that works out about four Bananas a Day to power your brain and this is calculated by the way based on a calorie intake that the brain needs so for context the fastest supercomputer in Europe I think it's called Lumi in Finland um it's been called exceptionally green and its power consumption is 8.5 million Watts so that's around half a million light bulbs well your brain uses just one so then the question is well what does your brain do with that one light bulb or four bananas apparently it does 1,000 billion calculations per second so so there's lots of other massive estimates out there uh this wasn't even the largest by several orders of magnitude estimates are obviously very speculative um but they're all massive and they all tend to be based on the number of neurons their connections and firing rates um but I think it's really important for the context that supercomputers can't actually yet match you know our complexity of skills um or the adaptability of the human brain so we actually Excel Way Beyond supercomputers when it comes to things like complex decision- making uh learning from experience so how does your brain compare to AI um so I mentioned that modern AI is already brain inspired however artificial neurons are highly simplified they don't they don't capture the complexity of biological neurons or networks like not even close um individual neurons are actually more like networks themselves and research suggests that modeling one biological neuron requires a five to eight layer di deep artificial neuron Network made of around a thousand artificial neurons this QR code should take you to the paper for that you have 86 billion neurons in your brain uh they work together to form a highly energy efficient low latency supercomputer that works just above room temperature of the equivalent of about four Bananas a Day so hopefully I've given you a sense of how amazing your brain is um as if you didn't know that already and how it's already been used to inspire the I guess fairly basic

AI that we have now compared to human intelligence so next I'm going to explain how key features of the brain are being implemented to catalyze our next generation of AI and Technology through the field of neuromorphic computing which is why you're all here so traditional vum and computers have physically separate Computing and memory units shown here on the left uh during computation data must transfer backwards and forwards like really fast so there's a bottleneck essentially for Speed and energy um whereas in neuromorphic architectures which are shown here on the right with the help of Dary Computing and memory occur in the same place so they're said to be collocated essentially individual neurons perform computation while memories represented by the strength of the connections the weights between neurons so the cospes so chips like this might be created with components uh like mistas for example which can emulate synaptic weights and this architecture improves speed it reduces energy consumption and what's really interesting is it enables massively parallel processing meaning that multiple problems can be worked on at the same time so this is particularly important this architecture for various use cases but also because as we reach the end of Moos log which is the number of transistors you able to physically make tinier and tinier to fit on a chip and it's also important because Humanity needs to massively reduce its energy consumption against the backdrop drop of creating ever more powerful AI so artificial neurons uh typically use continuous activation shown on the left they're always on while neuromorphic neurons they're said to be spiking so they're on or they're off which is shown here on the right so similar to sort of an action potential so the benefits for this are again power efficiency um and also applications where timing is important and this is given that their event driven so essentially they have um the potential for spatial and temporal Dimensions which then enables added spatio temporal encoding and processing of information uh you might be wondering bit about gpus um which also enable parallel processing uh research suggests that gpus are suitable architectures for deploying spiking your networks um which I think makes this a really interesting time for the field given how highend gpus are becoming ever more pervasive um so the brain learn strength of synapsis between neurons this is based on pre and post synaptic firing patterns I think David's talk will talk a lot we'll go into a lot more depth on this but there are many different typ types and patterns of this across the brain depending on the types of synapses such as excitatory to inhibitory excitatory to excitatory um and the neurom market field is working to leverage these rules um because of benefits for on chip learning and also applications such as pattern recognition and Edge Computing as well Edge Computing being quite a huge use case for neuromorphic computing

because of the sort of event driven nature and also the low energy usage and this QR code should take you to quite an interesting paper um on stdp that I found so what neuromorphic Solutions are available now you might just think it's s theoretical uh there are actually many different solutions out there which I'll just give you a really high level overview of so the human brain project has created several large scale neuromorphic computers including spica which is this one at the bottom this boards like maybe like I don't know the size of my face or something so that runs in real time um and it's comprised of multiple general purpose on microprocessors and there was also brain scales which is an accelerated analog architecture and it runs a thousand times real time so the um the board next to the blue one that's an actual credit card size um version of brain scales which they've recently made which I thought was pretty cool um and then there's also some big players in the space so this blue one here is Intel's Loi chip they're on to lii 2 now that's their neuromorphic chip and they have an open source software framework for that as well because they really want to catalyze the open source key Community to get involved with it so neuromorphic sensors also exist so this little blue thing in the middle is actually a neuromorphic camera it's maybe like this big um so they aim to recreate how our nervous system senses stimulate such as light so for example in a neuromorphic camera which is the one here each pixel works independently with a microsc resolution hopefully my GIF will work oh there we go so you can see each pixel working there which is pretty cool so compared to traditional digital cameras they have improved performance with motion and lower power consumption there was also a neuromorphic nose recently by Intel which was pretty cool um so it could learn the scent of a chemical after just one exposure um and then it could identify that scent even when it was masked by others and then finally this is a humanoid robot called an iub and what you can do is you can actually integrate neuromorphic sensors such as the camera and then neuromorphic chips maybe spa or brain scales into a humanoid device like this or other devices like a drone and then from that you can actually create embodied neuromorphic systems uh and this is something that we work on at the Smart interactive Technologies research lab in Sheffield in the UK this slide just highlights some of the potential applications of neuromorphic computing which I thought were quite interesting when you think about it so the understanding of context pattern recognition Advanced sensing few shot learning generalizing across tasks complex decision-making explainability and brain interfaces um so all these skills are really beneficial when you're thinking about human- centered realtime applications in Dynamic environments so things like self-driving cars for example and personally I think that neuromorphic systems um

are also likely to be the future substrate of brain computer interfaces probably a bit biased because I'm a neuroscientist but um they're low energy they're real time and they also have architectures which match our own Hardware so I do think we'll soon see the BCI field being catalyzed by neuromorphic systems particular maybe for hybrids of hardware and and wetwear so maybe even potentially containing people's own um you know brain cells which you can actually grow just from a hair cell and the particular focus of our work is designing AI which learns in a similar way to a human so it has an innate sense of curiosity and it learns through interacting with the real world so in the 50s Alan Turing said instead of trying to produce a program to simulate the adult mind why not rather try to produce one which stimulates the child's brain If This Were then subject to an appropriate course of Education one would obtain the adult brain this is very much the philosophy behind the neurodevelopmental approach um to Ai and neuromorphic Computing which I just wanted to highlight um there are some challenges in the field these are very high level but I'll just give you a little bit of an idea of it so training spiking neural networks is more complex than traditional neural networks also designing Hardware which actually implement spiking neural networks stdp on a large scale is said to be fairly challenging and then also developing algorithms which can actually effectively leverage all these like Technologies so the hardware the stdp it's an ongoing active area of research um so if you're here you're probably interested in active inference so I wanted to highlight this actually someone put on the Discord today one of these studies which is pretty cool so a couple of recent Studies have um combined neuromorphic Computing with principles of active inference so active inference comes from neuroscience and I would argue that it lends itself very well to neuromorphic architectures um in a recent paper on embodied neuromorphic intelligence so not not it didn't really mention active inference in it the QR code on the top right here um it was suggested that a real breakthrough in neuromorphics will happen if the whole system design is based on biological computational principles with a tight into playay between the estimation of the surroundings and the robot's own State and decisionmaking planning and action so some of those themes might sound quite familiar to people interested in active inference and U I would suggest that active inference is well placed to meet these requirements and I just wanted to highlight a couple of recent studies here so this one on the left gandal FAL recently demonstrated plasticity and Rapid unsupervised learning in a neuromorphic system using active inference principles the author suggested that their experiments could be adopted to implement brain-like predictive capabilities in neuro neuro morphic robotic systems and then there was the dish frin paper which some of you may be

familiar with by Kagan et al so this was a hybrid wetware Hardware neuromorphic system that the author's claimed was embodied the system showed rapid parent learning of the game of pong using the free energy principle for Learning and the authors claimed that the system exhibited synthetic biological intelligence so the field implementing active influence principles in neuromorphic systems is very new and the idea behind this stream series is to create a space and a community to share knowledge ideas and expertise to catalyze the field um I think some really exciting technological leaps are probably going to come from this area so thank you for listening to my uh quick run through neuromorphic 101 um and next up we're gonna hear from David so David over to you if you want to introduce yourself please thank you Sarah um I would share my screen [Music] yeah can you see now my screen the presentation yeah okay perfect okay yeah hello my my name is David Kle I'm a researcher and group leader at The Institute for Machine Informatics at the University of Bonn um so I'm uh leading the group on sustainable machine learning and we have a very strong focus on neuromorphic Computing that's why I'm here today and I'm going to start with a very similar motivation than Sarah did which was really a great um inspiration for this talk I think um so probably most of you have seen this um this interesting recent result and I don't mean Germany winning the basketball championship ship but um this really big leap in uh in artificial intelligence that we have seen in the last years especially the last two or three years so this is a picture generated from a prompt by a deep network and it's it's really amazing and was considered science fiction like only two three years ago and this was essentially made uh possible by a neuromorphic approach which is deep neural networks and this these uh deep neural networks have become huge now but this comes also with a caveat so basically the flip side is that this um among other problems maybe these models may have they consume uh huge amounts of energy so models like um DALL-E or GPT they as Sarah already mentioned they they would consume um energy budgets that are comparable to houses or cars so training um GPT a single time is like a giga hour approximately so that would be 300 tons of CO2 emission um and many times the which comes down to many times the life span of a of a typical car so this comes with two problems obviously uh this makes training these models only accessible to a very small number of very large players so essentially the big tech companies and secondly and maybe even more importantly this is not compatible with uh planet with uh limited resources so if the growth rate of AI continues like it did in the last years it will consume uh 133% of the global energy consumption by 2030 and it will basically outrun um the transportation sector in in another five years or so so this raises the question

does sustainable machine learning exist at all and obviously since I'm working in the group of sustainable machine learning I believe it does and why I think it does is because we know a system that um is very efficient and is still probably better than these these AI models which is the human brain which consumes as Sarah mentioned around watts uh or four Bananas a Day uh and uh so it's it's many orders of magnitude more efficient than the AI models we have today uh but so far we don't know essentially how these networks work um and especially how to train them and basically our our goal is to transfer now mechanisms from machine learning that's so you have this nice picture here uh basically we start from the machine learning site where we already know our way around um so we have these models that are wonderful and that are quite that that give us really impressive results but they are not efficient and we want to trans transfer them to um new efficient uh AI generation and our ideas is to use inspiration from Neuroscience um that make this transfer faster and and possible in the first place okay so and actually biology is a great source of inspiration and always comes around the corner with very surprising results and one of these results that I stumbled Upon A couple of years ago is that the the ability of um cospes in the brain so as you probably know neurons in your brain are connected for cospes um but if you look at the so so this is a paper from 2019 and they could actually identify individual Sy upsers and they could trigger them to uh make a synoptic release like a single transmission but if you look at this um um at these measurements you see that this is really covered in and noise so essentially if you basically average over this and zoom this out you see here these typical synoptic traces which is the average white line here but below that you see this huge chitter so there's they really like go several um standard deviations up and down um and this is actually very surprising given that uh neurons are probably the single most costly cell type in your body in terms of energy consumption they really in or in comparison they they they consume quite a bit of energy in your body so you would expect that these transitions or Transmissions that are communicated between neurons which are very costly uh should be highly reliable so this is very counterintuitive this results and has been puzzling neuroscientists for quite a while now um and then there is a there is a second um puzzling observ observation which is uh that the morphology of neurons looks some somewhat like this so this would be a typical pyramidal neuron you have in your cortex um but you see that this is actually for a cell quite quite big and elongated so this is this can be up to a millimeter in in the human brain uh which means that if a signups fires somewhere here uh it has a very hard time communicating with the cell body which is down here so electrical signals that are produced here uh May travel down here but the the signups up here has no way

of measuring uh the actual voltages at the cell body and this is actually the interesting place because here are the action potentials formed so if the the signups would really like to know about what is going on in the cell body so that it can make predictions about how the uh neuron will behave and how it interacts with the world and this is another very puzzling um um or open problem in in Neuroscience how this communication actually works out in in single neurons between the cell body and the uh and the signups it is known that actually the action potentials can travel back up so they see kind of this binary variable when the uh when the neuron spikes but they cannot actually measure the membrane potential down here so only the most prominent electric signals can actually back propagate through this uh and this this suggests that the signups actually has very sparse information about what's going on with this um uh in the cell body and uh most models of synaptic plasticity don't cover this at all so and we were wondering um how do this uh how does this interaction work how can the um uh signups produce useful learning signals given this sparse information about this important state of the of the neuron and our idea was that essentially these two observations these high levels of noise and the signups and this um large distance between cell body and and signups say which give these high uncertainties these are actually the two sides of the same coin so our hypothesis was that um actually we could use the same models um that we know already from from the behavioral level uh how an agent can in uh can act and perform in an environment of high uncertainty and we just apply this to um uh to every signups and oh there's an error so every signups should utilize this um these same models basically and uh this model would immediately suggest that actually synaptic Trans Transmissions should be noisy and these levels of noise would Express uncertainty about the environment and then we can use this model to derive um learning rules and we can compare them side by side to biology and this is the first thing I want to show you so um I just give a very quick introduction to the free energy model uh because some some of you might not be that familiar with it but this is essentially um a model to describe a situation like this you have a person that is interacting with some environment and here I assumed very simple so this person tries to um ball to some Target and we as humans we are we are good in solving such tasks and we are also good in solving such tasks if there is high levels of uncertainty in this so if the person may receive the some visual feedback but a lot of this feedback may be hidden so I you can imagine this going on behind some wall and the person now still might want to predict what is the trajectory of this ball flying towards the target so that it can make an correct uh action so we will assign some variables to this uh these states here so we have essentially this feedback that this person can

observe and we have this unobserved state of the ball flying here which we call U where the person doesn't have direct access to it would only see parts of it for example when the ball is at the corners of the wall appearing and then uh to model this essentially or to describe this uh Behavior the person would uh have an internal description of this trajectory so an internal model of this um State U uh and the model this model would then be updated to match the observed feedback and this can be described very nicely in this beautiful mathematical framework of the free energy principle so the idea is that you would essentially um establish a model of your internal States so essentially how the um internal States and the uh state of the environment interact and you would have a model of the feedback so how the states and the feedback You observe interact and essentially you can then write down um a loss function that uh that measures the distance between this model of the internal State and this model of the feedback and the uh external state um and uh then by essentially minimizing this distance between the two you can derive all sorts of behavior relevant um can solve behaviorally relevant problems for example learning but you can also use this for other things like uh figuring out what are good actions for example so making inference about both the internal States and the actions for example and this is shell the free energy principle and this object here happens to be what is known as the variational free energy which is also just coincides with statistical physics and this is where this framework has its name from but you see that it's all is probabilistic here so essentially you have two probability functions Q for the internal model and P for this interaction between uh States and observations and you have here uh a distance measure between them that you want to minimize so now if we look at the neuron and the signups and how they interact with each other we find a very similar picture so single signups um which we have here in green has um an internal state which we for Simplicity model only uh as the synoptic bait based on this the signups when it's triggered by a preoptic spike it would generate um a post synoptic current this would then propagate to the Som which is our uh external state which we cannot directly observe because it's too far away from the cups but we can see a feedback which is this spe propagating action potential which is this binary variable that tells us whether the neuron has spiked or not so this is exactly the same framework if you if we write it down like that and we can just use the same mathematics to solve it um so to solve it we only have to come up with a couple of of um we have to make a couple of assumptions so we have to uh write down a model for this guy here so this uh model of how the uh feedback and the and the external State interact uh but we have very good models for this this has been studied over many years so here you see how

typically a model neuron behaves so you have here um the membrane potential of a leaky integrated fire neuron and you see that this is just uh going up and down so this neuron would receive uh a lot of preoptic input and maybe also noise and eventually at some point it hits a threshold it would generate a spike so that would be the Z that travels to the downstream neurons and also back to the cups uh and then it resets uh right um but now this's uh so we can write this down mathematically is a very simple um differential equation but the uh neuron doesn't have access to this state again so it's again behind this wall it only sees this Spike events but we can actually for this simple case of really integrated F neuron we can solve this analytically so we can write down what is the posterior distribution of membrane potentials given the spike times and what uh comes out of this is actually a so-called um stochastic Bridge model or in this case of a uh of a leak integr in fire neuron it's an orang Bridge model so this can be written down analytically and this U I mean it's not simple but it's it's doable uh and we can then just use this directly so this model that we have uh to again write down this uh free energy functional so we make here an assumption how the signups actually produces post synaptic currents and how they're integrated in the in the neuron but that's also given by The Leaky integrated fire neuron and actually the stochastic um inputs that the signups generates so in this for Simplicity we only assume here basically gaussian Sy upses that would inject draw a gaussian random variable and inject this to inte great and fire neuron and then all these ingredients actually can be solved in closed form and we can derive learning rules that would minimize this pre energy functional now uh and if we do that um uh this has a bunch of nice properties because this Onin lbeck bridge is completely determined by the preent uh by the back propagating Action potentials so basically the times of the post synaptic um spikes that arrive at the neuron um this shape that we get here only depends on two neighboring postoptic spikes um which means that we get here automatically uh a learning rule that looks like this so learning rule that only depends on the difference between two postoptic spikes which we call here ΔT_2 um and uh the difference between the post synaptic Spike and the actual input that is triggered at some point uh on the preoptic side uh and we can basically make here this lookup table and just uh just compute what would be the update that these signups would need to make so that it learns optimally in terms of this free energy principle and this is the shape that we get out you see that there is a strong dependence on the post synaptic firing rate but there's also a dependence on basically this typical STP that Sarah mentioned before uh what is the relative positioning of of the pre and postoptic Spike so in a nutshell this model can now be split into essentially two Pathways so we have this atoc

response which basically just whenever there is a preoptic spike that triggers an action in the cups he would draw from this scussion distribution and inject it into the um into the neuron and then there is this postt talk uh update where we where the signups would look up um in this orang and wenck Bridge what would have been the optimal um output that it should have generated so what would have been the optimal action and then it compares the actual action with this optimal action according to this free energy principle and then generates a delayed response response which is an update of the synoptic weight okay and importantly this uh internal model is only implicit it's encoded so to say into this um uh Spike time dependent plasticity rule uh so how do these rules then look and do they compare to how do they compare to biology and uh actually they the fit is quite nicely given that this is derived really from first principles without making any assumptions so this is the measurement in biology this is this um B at Al rule B and po very old work where they actually did this in uh in vro studies where they measured the injected pre and post synoptic um spikes and then they measured what is the weight change in the cops and this is the rule that is predicted by our uh model and you see that it at least in a first order approximation it gives us very similar shapes and this also makes sense because um the the signups wants to um change the most when it's close to a uh close to the pre and postoptic Spike times because this is where or or the postoptic spike times because this is where it knows the most about the state of the PO neuron the free energy principle would actually suggest uh this kind of cones U um with almost no assumption essentially but uh we also get because we have a not just the first order uh Spike time dependent plasticity rule but uh we also have this dependency on the postop firing rate we can also compare this to other results and this here is this old by gr and Brunel so this is actually a model but that was very detailed um describing the um the uh plasticity based on the pre and postoptic firing rate in uh in a signups and this is what our model predicts so if we inject random pre and post synoptic popik tra uh trains with different rates our model would predict this shape which again is not a perfect match but given that this is a u very idealistic model it's actually least the the main features that uh low firing rates on the postoptic side would lead to depression and higher to potentiation um are reflected in this okay uh I I assume I still have 10 minutes right okay so I would um give a quick intermediate summary and then I want to show some other work where we actually apply this now to um to actual machine learning model um so what we have seen here is that um Sy upses are actually um are very stochastic and it was a very this was a big puzzle and we suggests that actually the um uh that the the synaptic noise is actually the signups is way of um reporting its own uncertainty about

the environment where the environment is actually the um the POS synoptic neuron and it really interacts in this fashion of the free energy principle with this post synoptic neuron or that's a very nice way of describing it and uh if you're interested more there's a a paper a pre-print out you can read up on all this okay um so how does this now connect to neuromorphics and actually so we are not actually doing neuromorphic Hardware we are doing doing neuromorphic algorithms so we try to to bring these Inspirations now into actual machine learning models and we thought that this might be a good um attack angle to um solve a problem that is well known in machine learning so I I just um uh yeah Drew here a very simple um convolutional neur networks with various convolutional layers and then maybe some dense layers that you would have in your machine learning algorithm and the way this is trained as many of you know I guess is through um end to endend and arrowback propagation so the idea is that you have a training set which has inputs and uh targets so for example in a classification task this could be uh pictures of cats and dogs and you would have targets which are U class labels so to say so some there's there's actually artificial neurons in there and some one of these neuron may be active for cats and one may be active for dogs um and in your training that data you have exactly these labels um that were generated by human that were sitting down doing this by hand and then during training you show this examples to the network by propagating this inputs uh all the way from the input layer to the output layer then the output is here compared to the uh to these hand label targets and then the mismatch between the two is back propagated through all these layers back to the input and all the weights or the synoptic weights that are here uh in between uh inside this layer would then be updated accordingly so that after doing this many many times uh this network becomes good in telling a part C dos so this has a problem this this algorithm it works great in practice and it's it's the foundation of all these models we have talked about like D or J gbt but it is quite inefficient and the problem is this what is known in the literature as the Locking problem so if you would split up this network now into blocks which I already did before but this is arbitrary but for for implementing this efficiently in terms of a software algorithm it might be interesting to do that and now you would want these blocks ideally to to run in parallel so that you can basically uh show the first example and uh on this first block and then already train it while the second um block is doing something else but this is not really possible with um end to end back propagation because of this locking problem because the activation of the second block depends on the activation of the first block so you have to propagate it all the way to the end then you compute this error and you would then back propagate and only when this is

done uh you can start the next Epoch where you show a new bunch of um of examples uh and you see that during all this time here the the threat that would run this first block maybe uh would be idle and has to wait essentially all the time and this obviously makes them very inefficient and now I our idea was that we use basically uh what we had learned from this earlier model on um on how sign ups communicate over these long distances for this free energy principle and also apply it just to a deep neuron Network and the ideas that you have here again this basically you have already this um this generation of inputs to some output but what what is missing to make it applicable to the free energy principle is this feedback that you always need the idea was that we put here very lightweight um uh feedback Network so essentially each of these blocks now in this deep neural network would uh be accompanied by a feedback block uh that locally generates a Target so we used to really in the simplest case which we have so this is very recent work we only used linear blocks so far so these are single linear layers and we would generate now these these outputs here in these feedback blocks and then use the free energy principle to derive um uh a local loss that allows us again to minimize both these uh feedback weights that we have here and also the weights in the forward Network um so it's essentially the same idea so we have here these outputs which we interpret now as a parameters to a probability function so we can apply this probabilistic framework uh but now the all the rest basically rolls out the same way so we uh assume that these outputs are essentially um the internal states of this model and we have this um uh this observations given in the uh inputs and and the targets and now we try to minimize basically uh P would now be this feed forward Network and Q would be now a function that uh contains um features of both the feedback and the uh and the feed forward Network and the nice thing is that if we um uh I don't have time to go into to the details now but if you write this out you see that um this actually uh decomposes this lock term here decomposes into local linear terms that give you these local losses here so essentially that you can minimize here block local between B and the corresponding feedback block um a loss function and you can then actually do this in parallel so because uh maybe the picture is good to see here so what you have to do now you have a bit of overhead because you have this feedback block so this would be the two uh uh execution times of the uh feed forward block and the feedback block but in principle they can run in parallel uh and once the forward block is done the next forward block can start propagating through this network um but simultaneously already the forward block because it already received a Target here can start uh updating the weights and when it's done it's free to operate on the next Epoch so there is no locking anymore in this in this framework okay so I'm

pretty much done I'm also out of time I think so how does this perform of course we changed the the learning algorithm now we have to also go back and see if this is uh still uh giving us the same performance and actually for um so this as I said is this is the first um results we have here now and at least to um midscale uh data sets like Cypher 10 or so this seems to actually uh perform very well so we are tried it for for standard architectures fashion mnist with resnet 15 and resnet 18 we worked mostly so far uh for like small data sets like fashion mnist we we've applying free splits to resnet 50 we get basically the same performance as standard backprop as networks get deeper uh you see that there's actually um the our problem that we have now is actually overfitting because we have these local targets it seems that these smaller blocks actually overfit to some extent this is not so severe for still up to tasks like Cypher t so we get quite close already but if you go now for really large tasks um there's still something missing we are uh for for like single splits we're getting there but we're not reaching all the way up to back propagation but it's still interesting to see that you can apply this um this principle also uh to this standard machine learning algorithms okay this is my second summary so actually we found that um deep Neural networks are surprisingly good in generalizing over probability spaces this is how actually this work started um and we we our idea was to exploit this um and to utilize it for to distribute learning in the in the same fashion as um in the first project I showed you um and to solve this credit assignment Problem by generating these feedback networks um yeah that's basically it and then um I want to acknowledge my my co-workers and my students so I have two very good um um PhD students KH and Cabel who is now here in in Boston and works um on this topic and I the first um project I showed was B I did together when I was in grad school with Christian TL and uh this project the second project is I work I worked closely with Christian Mead and Anan Suprun and yeah thank you thank you very much David that was absolutely fascinating I think it's incredible how closely the sort of model matched biology like considering you derived it from first principles I think that was really cool um I did have a question just about the the last sort of the bit you spoke about the convolutional network and you know you said traditionally has to go all the way end to end which is really inefficient and then you showed the results that you got did you look at energy consumption with with yours as well uh not yet so we have uh we are actually currently working on a so this um it's actually not so easy to implement these things in in standard machine learning uh toolboxes we have a so Kal is currently looking into this uh the PHD student in Dron um he has an implementation now and he's now evaluating how uh how well we can make use of this par titioning in practice but we are actually quite

confident that this um that's for paralyzing it uh it should be there the question is how much you save in terms of energy because um for these smaller scale models that we use now reset 18 reset 15 the the effect might be not that huge so once we ramp this up to really larger models the effect should be bigger but yeah this is ongoing work very cool thank you um and then I was just wondering as well like this this local error back propagation is that something that other people have tried with these convolution neural networks or is this quite a new way of implementing it uh there is a bunch of approaches that do this so uh there is for example I mean the closest I guess is Target propagation which has been proposed which essentially uses random feedback weights uh to back propagate here so these these guys would not be trained um and those as far as I know this works this works nicely also for small scale um uh problems but as far as I know they don't perform that well even for for Cipher 10 they they it already starts breaking down because these random feedback weights are just too coarse an approximation I think and this is the first um okay maybe I have to be careful I think this is the first um method that allows you to train this feedback weights that is not a contrastive method so there's a bunch of methods that use a contrastive step um so you have maybe seen this forward forward algorithm and and all these things but what they have to do always is they send in uh uh they send in the actual input data and then they send an kind of anti- input so an anti- input that is uh usually generated artificially so they do some Distortion to the input to make it uh and then they train the they have to use both informations locally uh to do the update so the network has to keep in memory uh the input and the anti input and the the responses and this makes um this makes this approaches a bit harder to parallelize in this and the nice thing here is that we derive an uh an upper bound to this uh variational free energy loss that can be spelled out completely by forward propagation and that I think is new so that is the the new bit of of this very that's awesome well yeah few few comments one piece that kind of between the two of your talks that was at least a new distinction to me which was is the difference between the neuromorphic hardware and the neuromorphic algorithms so it's not just about new hardware or wetwear though that would be great to see it's almost like there's this intermediate or a bridge step with using algorithms on the hardware we have today that like Sarah mentioned the spiking neural networks which are amenable to gpus or just using standardized um CPU multicore scheduling approaches you can already do more with what we have using the neuromorphic algorithms so it's not just um a material science topic but also there's a lot at the really micro scale that we can learn related to noise processing scheduling and then also even at higher levels of abstraction probably learning from biomimicry and

cognitive systems more generally but that was like that was a distinction for me yeah so maybe to to add um yeah so I think that the problem becomes really now more pressing as these uh neuromorphic uh devices become AI the hardware devices become more mature and they usually cannot really shine on this uh on these standard machine learning algorithms because they are really optimized for for gpus so you need to think a little bit so you have to take one step back and think again about the algorithmic side um to to really use them to to full capacity and we also as you have seen that we collaborate with um Professor Maya who is doing this Spinner chip in Dron but there's also other approaches and like the L chip that Sarah mentioned from Intel and so on um and they are really looking into this now they also from the algorithmic side I I find it really useful it's kind of like a mindset or a mental framework when I'm thinking about computers or AI and I this is probably because I'm a neuroscientist but I also have to translate it to well how does the brain work how does the information processing work etc in the brain and and when I came to sort of computer science and AI after Neuroscience I found myself naturally translating it but I feel like the framework is is just a really useful way of understanding comput at the end of the day because our brains as I said are just these massive supercomputers and I'm constantly reading papers on computer science or whatever and and then once you you can conceptualize anything really as well how close how neuromorphic is this and then if you start thinking about well how how could you tweak it so it's slightly more neuromorphic and is that then going to give you these gains that we get with the brain like is it going to give you some like extra parallel computation or is it going to give you some Energy Efficiency so yeah I I find it NE like the definitions when I was looking at the definition is I think it really depends your rights a definition and because it's such a a dynamic area at the moment as well um I think it's going to be it has been changing and it will be changing but for me I feel like neuromorphic is more of a frame like a mental framework where I look at things through conceptualize yeah I think it's it's also not very well defined I mean uh in you also mentioned that that actually artificial neuron networks are a neuromorphic uh concept if you want and they were from the first day and it's actually it's a it's a big success story right if you look into the 90s or so when these support Vector machines and and these these alternative models came up but none of them have have outlived the the new neuromorphic approaches so uh it's actually very nice but still there is this communic community that thinks that there is more features from the brain that you need to put in to uh to get to the real thing yeah so I think this is a bit of a it's not a very well defined term actually and I think with your research yours is almost like the smallest level that I've seen

people look at it on I don't know if you've seen anything else but we're not just talking about like a cell level of free energy and active inference we're actually talking about like a cell structure and then like so then you think well how small does it go are we going to talk about cell subcellular structures you know eventually like mitochondria using free energy in a similar way with compound compartments and yeah so I guess actually' be interesting to get your thoughts David on you know you talked at the start about how sign ups are compartmentalized um do you different different sort of instantiations of this in different compartments just in within one signups almost like are you wanting to sort of look at that granular level or is it is it more now taking what you've learned from this and putting it back into how can we sort of you know make the AI more efficient uh yeah we are going much more in this direction now that we see how we can build this back into um AI models um so I think the the free energy so one has to be a bit careful when using the free energy principle because it's such a powerful General framework that you can apply it to basically anything um and it's not necessarily you will come up with a useful result in the end just applying just putting this um and we basically I mean this started actually as a side project this was my kind of uh covid um pandemic lockdown um project and I was just curious about this and um whether you can actually solve this because I thought the signups is maybe simple enough that you because when you when you go into the papers um they have at some point to go into some approximations they do some um meanfield usually so they go for first modes and then you can solve these guys for more complex even for neurons it's hard actually if you go to the neuron or network level it's it's hard it's it's very involved math but for a sinus is simple enough actually so you can actually do this and and spell everything out if you make the right assumptions and um and really just derived these things and that was kind of an just an kind of a game I I I went into and then it turned out to to work quite nicely I think in the end yeah very um I'm not sure if you I like mitochondria or so I'm sure you could apply the same principles but I'm not sure if the results you get would be any meaningful or would help you in any way well that's that's always the the risk you you invest so much time and then in the end you get some results and you don't know that was also something that I thought was quite interesting which was the synapse was the agent it's really easy think oh we'll make a agent-based model of a neural system first off that tends to not include Glia or non- neural cell types but it's almost like a doubly unquestioned assumption that the cell would be the agent but then it was a great transition from the person throwing the ball over the wall that's an action Centric approach where you only have partial visibility of the consequences

and then that is the exact scenario that the synapse finds itself in in a different way or it could have been set up so that a neuron is the agent we're building maps not territories and so then just like you said free energy principle it's a principle for everything and so just making principled statements about things is table stakes and then I guess my question for you is then what does make it useful or in your learning and tinkering around with these models what differentiated situations where you applied free energy principle or active inference and you felt like it was providing a contribution to your research Direction versus where you played around and it was like well that was too logical um I think it the free energy principle makes sense in a context where you have um incomplete information so as soon as you so for example in the signups case right the question we started with was um this problem that the signups has to solve that it has incomplete information about the um the state in the cell body um because it only sees this uh kind of or that's the Assumption at least of the model and and also what we get from the experimentalists that they essentially only sees this spe propagating action potential so sees a single binary variable about um about the state of the of the Soma so essentially this is a problem of uh of incomplete uh information and also the second ingredient that you need an agent that you need some form of agency I think if you apply the free energy principle to a a system without agency so if something is not interacting with an environment in a closed loop uh then it becomes really sketchy and I think already this model is on the edge when it comes because these these models don't really have an agency but they at least produce an output right so you can still think of this as an as an interaction with an environment but some as soon as you lose that I think then there would be simpler models that can just give you the same yeah actually in the signups in the signups case the the agency is only this uh this adding noise actually in the model because the signups is triggered PR synoptically and then it it adds this it uses its internal state to add the right amount of noise which is probably already the minimal agency you could imagine Sarah you want to ask a question or I can ask a question um yeah I was more just a comment like I think it's quite interesting like people talk about biology some people say it's not real science because it's all messy and noisy but I think your it works really interesting because it's like you say the synaptic noise is actually a reporting of uncertainty so in that Essence it's actually probably quite accurately reporting and and the messy World rather than the biology itself just being all messy but that's just what I was thinking about um yeah I think I was curious as well like you said this was like your lockdown project but I'm just interested in how you sort of came to the free energy principle how you came across it was it something you were quite familiar with

already or or some of your network or peers were talking about it or did you stumble across it in a paper um I mean I was when I was during my PhD I was interested in um in variational methods and and probabilistic methods um and then I started reading about this and so I read a bunch of um kristance papers and I I found this interesting and um actually my my PhD supervisor always um encouraged me not to not to go in that direction uh and then after I finished my PhD I thought okay now I I can do what I want I try it out so yeah and then I I guess do you think it would be worthwhile like next steps for you or for the the field actually trying to implement this on you know maybe some of the more analog chips that are being built in the space like the analog neuromorphic chips which I know you can have like some synaptic Dynamic synapses and and things like do you think it' be worthwhile um trying to implement own hard Hardware or what your thoughts on that um I mean the triplet rule that comes out from this first work I showed I think that would be interesting to implement it um it is the nice feature is that it it should in principle be kind of this should have this self-stabilizing picture because it's really mimicking the uh the Dynamics of the the membrane of the cell membrane so if the if the neuromorphic hardware would uh so if if the the model in the signups and and the Neuron model match up very well the model should uh give you this nice uh self stabilizing feature so that neurons really not go into some epileptic States or so and you get this for free from this model that's what we saw in the simulations at least but um in the simulations of course we we had full control over this Dynamics matching up in the right way so um that is probably a bit more tricky for Hardware but it's probably solvable so it would be interesting yeah then what you get for it is that you have this purely event based tools right which only use um prepost spikes which which is nice very cool thank you yeah Daniel if you had a question well that's a great principle there which is like if you can design the neuromorphic algorithm so that it harnesses a material feature like the actual leaky permeability of a membrane or actual spatial proximity if you can leverage a material feature analog feature that isn't virtualized then it's already in adjacency into future Hardware so that's one great point and then to s's point about like almost biology not being a science which um there's a famous quotation there will never be a Newton for a blade of grass because some people say yeah it's a different biology is more like history because whether you approach this from a development or ecology or Evolution perspective biology is a historical science it's not like a real science and then that reminded me of the cross country shirt that says our sport is your punishment so it's like well no like that your noise is biology's signal and that's how it happens my question was about this tension between I guess neural and computational ways of

looking at the resources associated with computation so from the Von noyman Paradigm we have a lot of shared reference points CPU Cycles Ram capacity and all these kinds of things and like even in your introductions you conveyed like well this is how many CPU Cycles it's going through or this is how many parameters would have to be stored or something like that however that's referencing another Paradigm so what do resource descriptors or capacity descriptors look like when we're outside the space of okay yeah power consumption that's something that you can put into a box and just use a bomb calorimeter that's kind of like a low hanging fruit but now okay Beyond just the sheer energy or caloric requirements what can we say that is like analogous to the way that we talk about the processor the Ram or the hard drive on a computer I mean I'm yeah I I'd have to think about that one some more I do think there was some interesting comments in the paper on that slide I showed that talks about um the brain and energy there was a paper I linked to I'll have to get the reference and let you know what it is because QR C's gone now um but that had some interesting ideas I think on what you're getting out there but I yeah I'd have to defer to the paper I mean how do they describe what is being designed do they say it has this many of this type of component and then that might do nothing though so how do they describe or evaluate these different designs or algorithms I think it's all different depending on the use case like that's what I've found really like the language is is different depending on you know if it's written by someone maybe with more of a neuroscience background or engineering background and then you kind of you said that some terms they're more interchangeable than others but I do think the terminology is something which needs to be looked at a lot more closely in the space because then I think that will help everybody working in it to be on the same page a little bit closer cool well any other thoughts or questions David first and then Sarah also I'm very curious what direction will this series go but first David what are any other kind of closing comments or directions you want to provide um not really um I mean I would say thanks for for having me today was really was really a pleasure to discuss with you thank you David it was amazing to have you on I think your work's absolutely fascinating and I think it's going to be have lots of benefits in the future for implementation which is always nice to see as well um yeah so what was the question where where do I see the series going um hopefully we can have a new guest each month um I think it'd be kind of cool maybe next month to do um to have someone who's building Hardware so like maybe someone from the brain scales team or spica team or something like that would be pretty cool um but yeah really I just want to have a space for people who are interested in this intersection to uh meet people who and see talks and you know Reach Out

people who are also working in the space because um it's pretty Niche but I think it's pretty important um actually having said that David could you let everybody know like if they wanted to reach out to you what's the best way for them to do um I'm not very active on this Discord channel so maybe email is still the best uh to reach out to me I guess cool do you want to give your email just oh I think my email should be easy enough to find but you can also get the email out there short people can check the papers and then in the active inference Institute Discord there's the neuromorphic Channel all right thank you David and Sarah really cool to see morph stream kick off its developmental trajectory this way so till next time thank you bye fine